Next experiment: tune the base and oracle together

Date: 2026-10-16. Status: recommendation and draft experiment; **no new run**. Scope: the prime-field factor-base study imported by PR #1898. This is a research handoff, not a manuscript or a new performance result.

Recommendation

Use low-x as the historical control and optimize **base size plus decomposition cost** in the existing native IC pipeline. Keep multiplicative-coset/Kummer bases as challengers. A useful base supplies independent relations cheaply enough to reduce the full cost to a verified answer; coverage alone does not decide that cost.

The next bounded experiment is the draft XP-20261016-prime-base-oracle-tuning. It is not ready to dispatch: the exact native baseline, executable spec, inputs, budgets, checkpoint and independent replay command must be frozen first.

Evidence behind the choice

The historical study, preserved in PR #1898, found low-x fastest in its 24-bit exact-join cohort. The coset/random throughput interval includes no difference. Its algebraic norm solver was a Sage prototype and its join was native C. Those stage CPU comparisons do not establish a native pipeline ranking, and neither their timing conditions nor their cohort choose the best current repository pipeline. The original report, data and hashes stay intact.

Read-only checks of current source reports clarify the possible follow-ups:

- The generic-field F4 study already runs small Kummer systems at P-224 and Dickson torus systems at P-192/P-256. A wide-field port is no longer the first task. Its matching solving degrees concern the recorded small systems, not a full DLP at those sizes.
- The msolve degree study reports productive degree-8 rounds at four-point decomposition, N=20, in its raw presentation. The degree-7 plateau therefore failed for those runs. The reduced native presentation remains a distinct unresolved check. This is not a proof that every tower algorithm fails or a claim that the native cell has completed.
- The charged isogeny-tower study found no benefit over a plain matched-size base with its table oracle. This supports measuring the base/oracle combination, with construction costs included.

The encyclopedia and backlog remain useful indexes, but their older statements that no NIST-prime tower system has run, or that the four-point plateau is entirely untested, need these source-report qualifiers. This note does not modify their frozen reports or close their backlog cards.

Ordered work

1. **Freeze the native incumbent and establish correctness.** Use existing Rust ecbench components. On each registered prime-order curve, identify the lowest-cost currently verified native IC configuration and preserve its code/spec/input/receipt hashes. Low-x remains a mandatory historical control even if another native configuration is better. Cross-check the native decomposition outputs against exact small-group solution sets and the archived identities. Do not extend the old Sage execution path.
2. **Screen base size and oracle jointly.** Sweep a preregistered grid around the incumbent base size using low-x and the admissible coset/tower base; use random bases as controls. Compare actual signed point classes and memory, and record lookup-table size separately. Match actual base sizes for family comparisons; when optimizing size, explicitly report the size change. Freeze the arity in each comparison. Start with existing exact native pair-table/subtraction oracles and assess batching or table layout only where the code audit shows

that work has not already landed. Measure verified independent rank per charged work, rank stalls, setup, failed attempts and memory as stage diagnostics. Use disjoint tuning and fresh holdout curves/targets; the already-inspected archive is a regression set, not a fresh holdout. No target scalar labels may select a base.

3. **Validate the selected recipe end to end.** Freeze the selected parameters before holdout evaluation. Run incumbent and candidate on identical, separate one-target workloads through recovery and independent point verification, with matched rho on the same targets. Include cold setup, oracle work, failures, filtering, verification and linear algebra. Report online cost separately; do not divide a multi-target batch to invent a one-target cost. Independently replay the sealed records before promotion.

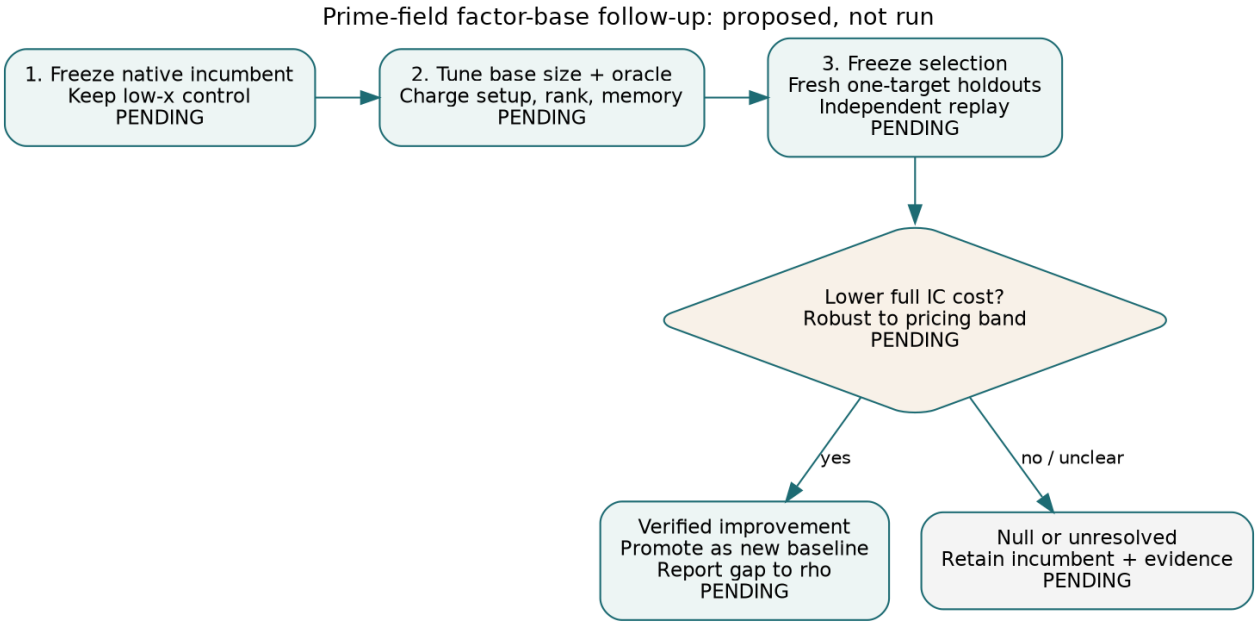


Figure 1: Proposed decision sequence; every execution step is pending

Editable diagram, SVG, PDF.

Decision and accounting rules for the future protocol

Use the repository's counted cost S, with an explicit inventory of every unpriced native counter on every arm, including rho. Emit missing counts before ranking. Apply the same native-operation weights 0, 0.1 and 1 GAE to every arm, and show pinned calibration as an additional point only when available symmetrically. A ranking that reverses inside the band is unresolved.

The acceptance target is a verified reduction from the frozen IC incumbent on fresh matched workloads, robust throughout that band. An engineering improvement counts as success even while IC remains above rho. Report both baseline/candidate and distance-to-rho ratios, sample sizes and paired uncertainty. Use the repository isolation and A/A requirements for any wall claim. Preserve caps and unsuccessful runs; a timeout is not a win or a nonexistence proof. A stage improvement remains a stage improvement until it reduces the full pipeline cost.

If holdout improvement is unresolved, retain the incumbent and report the bounded outcome. The existing heavy PKM degree card remains a separate structural workstream. Reconcile its raw/reduced-presentation question with the msolve evidence before requesting a large-memory run; neither that question nor larger NIST-prime coefficients is a prerequisite for a useful measured constant-factor improvement to the current pipeline.

Status and reproduction of this handoff

Deliverable	Status
Evidence-linked recommendation and diagram	Prepared in this change
Draft experiment card	Prepared; not dispatchable
Exact native incumbent and runnable protocol	Pending
Tuning sweep, holdout solves and independent replay	Not run

Pinned source-report revision on repository main:

`f334da9fea56bcd1480cd198d238a5e36de36108`.

No new curve, theorem, benchmark value, asymptotic claim or pipeline ranking is introduced. No Lean proof or algebraic certificate is claimed. Historical and native CPU results are not combined. New measurements, if run, need their own reports, benchmark copies and affected dashboard updates.

The current root AGENTS.md sections 7a-7c were checked for applicability: the scoreboard, progress timeline, leaderboard and lab browser remain unchanged because this planning change introduces no new measurement, identity, replay or corrected end-to-end ratio. Existing study plots are linked, not regenerated. The diagram is qualitative and marks every execution step as pending.

Run `sh build.sh` in this directory to rebuild the SVG/PNG and PDF from the DOT and Markdown sources (Graphviz, Pandoc and XeLaTeX). The PDF is checked page by page before publication. Conductor, Cargo and the backlog executable are absent in this workspace; their attempted commands returned 127. The referenced `.conductor/WORKFLOW.md` returned 404. Structural card checking uses the committed JSON schema here; native `backlog validate` remains a CI gate and is not reported as passed locally.